

Saad Hossain

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EDUCATION

Bachelor of Applied Science (BASc) in Biomedical Engineering
University of Waterloo — Specialization in **Medical Artificial Intelligence**

Sep. 2020 – Apr 2025
GPA: 94/100

PUBLICATIONS

- [1] **S. Hossain**, T. Tseng, P. Pandey, S. Vajpayee, N. Nonta, M. Kowal, S. Simko, S. Casper, Z. Jin, K. Pelrine, S. Rambhatla. TamperBench: Systematically Stress-Testing LLM Safety Under Fine-Tuning and Tampering. *ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*, 2026. [\[Link\]](#) [\[GitHub\]](#)
★ Oral presentation (top 4.5% of papers).
- [2] K. Kaai, **S. Hossain**, S. Rambhatla. Bringing Equity to Classification: Domain Generalization for Domain-Linked Classes. *IEEE Transactions on Pattern Analysis and Machine Intelligence (T-PAMI)*, 2025. [\[Link\]](#) [\[GitHub\]](#)
- [3] R. Chaudhary, J. Ho, D. Smith, **S. Hossain**, J. Hargun, B. VanBerlo, N. Murphy, R. Prager, K. Rikhraj, J. Tschirhart, R. Arntfield. Diagnostic accuracy of an automated classifier for the detection of pleural effusions in patients undergoing lung ultrasound. *American Journal of Emergency Medicine*, 2025. [\[Link\]](#)
- [4] **S. Hossain**, S. Vajpayee, S. Rambhatla. SafeTuneBed: A Safety Assessment Framework for Harmful Finetuning Defenses. *International Joint Conference on Artificial Intelligence (IJCAI) Workshop on User-Aligned Assessment of Adaptive AI Systems*, 2025. [\[Link\]](#) [\[GitHub\]](#)
- [5] C. Liu, **S. Hossain**, C. Thomas, K. Lai, R. Vemulapalli, S. Rambhatla, A. Wong. LangDA: Language-guided Domain Adaptive Semantic Segmentation. *Advances in Neural Information Processing Systems (NeurIPS) Workshop on Adaptive Foundation Models*, 2024. [\[Link\]](#)
★ Best Vision Paper Award at CVIS 2024.
- [6] J. Park, K. Kaai, **S. Hossain**, N. Sumi, S. Rambhatla, P. Fieguth. Domain-Guided Spatio-Temporal Self-Attention for Egocentric 3D Pose Estimation. *ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*, 2023. [\[Link\]](#) [\[GitHub\]](#)
- [7] K. Kaai, **S. Hossain**, S. Rambhatla. Are all classes created equal? Domain Generalization for Domain-Linked Classes. *Advances in Neural Information Processing Systems (NeurIPS) Workshop on Distribution Shifts: New Frontiers with Foundation Models*, 2023. [\[Link\]](#) [\[GitHub\]](#)
- [8] J. Park, F. Barnard, **S. Hossain**, S. Rambhatla. Implicit Stylization for Domain Adaptation. *International Conference on Learning Representations (ICLR) Domain Generalization Workshop: What do we need for successful domain generalization?*, 2023. [\[Link\]](#)
- [9] J. Park, K. Kaai, **S. Hossain**, N. Sumi, S. Rambhatla, P. Fieguth. Building Spatio-temporal Transformers for Egocentric 3D Pose Estimation. Joint International Workshop on Egocentric Perception, Interaction and Computing (EPIC) and Ego4D, *IEEE/CVF Computer Vision and Pattern Recognition Conference (CVPR)*, 2022. [\[Link\]](#)
★ Oral presentation.

EXPERIENCE

Machine Learning Developer

Deep Breathe

May. 2022 – Present

London, Ontario, Canada

- Developed AI screening tool confirming with 96% certainty the presence of fluid in lung ultrasound scans.
- Spearheaded integration of **V7 Darwin** labelling platform into internal data pipelines and ingestion workflows.
- Leveraged **PyTorch** to auto-track objects from **Labelbox**, cutting labelling time per video from 10 to 2 minutes.
- Reduced dataset prep time by 91% deploying **Python** & **Bash** pipelines on **Azure**, generating 7.8M M-Mode images from 140K B-Mode videos per version.

Machine Learning Research Intern

Kolena Inc.

Jan. 2024 – Aug. 2024

San Francisco, California, U.S.A

- Fine-tuned Vision-Language models using **HuggingFace** & **PyTorch**, cutting GenAI review time by 300%.
- Accelerated text dataset creation by 15x by developing LLM-powered web-app with **Streamlit** & **OpenAI SDK**.
- Raised financial data accuracy by 8% by refining PDF-parsing & LLM-prompting in RAG with **LangChain**.
- Extracted metadata features for Kolena's AI Quality Platform, enabling CV and NLP users to stratify test datasets by subgroups and identify edge cases using **Python**, **OpenCV**, **NLTK** & **SpaCy**.

Software Engineer

Jan. 2022 – Apr. 2022

IntelliSports

Montreal, Quebec, Canada

- Used **Scikit-learn** & **SciPy** to feature engineer IMU Sensor data, boosting cheat-detection accuracy by 30%.
- Elevated mission count by 2.3x by deploying detection pipelines on **EC2** for punch, crunch, high-knee, & jump.
- Identified onboarding flow with +50% retention by conducting decision-tree & SVM analysis on early user data.
- Built data visualization platform with **PHP** & **MySQL**, showcasing user physical activity growth of 300%, employed in pitches that secured \$1.4M in funding.

Student Researcher

May. 2021 – Aug. 2025

University of Waterloo — Apple, Nissan

Waterloo, Ontario, Canada

- Engineered open-source toolkit benchmarking safety of 38 LLMs under fine-tuning with **Transformers** & **vLLM**.
- Built infrastructure for algorithm that improved tumor-detection accuracy by 9% on data from new hospitals.
- Outperformed egocentric pose estimation baselines by 38% via insertion of spatio-temporal Transformer models.
- Used **ROS**, **Flask** & **PostgreSQL** to deploy facial identity & emotion recognition pipeline on the Reem-C robot, highlighted by researchers in international robotics conferences.

HONOURS AND AWARDS

- Alumni Gold Medal *Finalist*
- Best Capstone Design Project - Biomedical Engineering *\$500*
- CRA Outstanding Undergraduate Researcher Award - *Honorable Mention*
- Best Paper Award *Conference on Vision and Imaging Systems 2024*
- Yuen Family Foundation Award for Final Year Design Project *\$10,000*
- President's Research Award *\$1,500*
- Engineering International Student Scholarship *\$10,000*
- President's Scholarship of Distinction *\$2,000*
- Term Dean's Honour List. Fall '2023 & Winter '2023 & Fall '2021 & Fall '2020
- Top AS Level Exam Scorer in Country *Cambridge International Examinations 2019*

PANELS AND MEDIA

- Translational Pathways in Biomedical Engineering: From Lab to Clinic. CUBEC. *Panelist*, 2026. [\[Link\]](#)
- Featured in News Article: BME undergraduate student received Honorable Mention for Outstanding Undergraduate Researcher Award, University of Waterloo SYDE News, 2025. [\[Link\]](#)
- Augmented Reality for Guided Lung Ultrasound Acquisition. Ibtikar Seminar Series. *Guest Speaker*, 2025. [\[Link\]](#)
- Enhancing AI Testing : The Power of Metadata Hydration. Live Webinar Hosted by Kolena, Inc. *Invited as Key Speaker*, 2024. [\[Link\]](#)
- Mastering RAG Systems for LLMs: From Pitfalls to Performance. Live Webinar Hosted by Kolena, Inc. *Invited as Panelist*, 2024. [\[Link\]](#)
- Crash Workshop on Physiological Computing, Univeristy of Waterloo. *Presenter on Inertial Measurement Unit (IMU) Feature Extraction*, 2022. [\[Link\]](#)

TECHNICAL SKILLS

Languages: Python, C++, C#, MySQL, PostgreSQL, PHP, HTML

Tools and Technologies: PyTorch, TensorFlow, Sklearn, Git, OpenCV, ROS, EC2, Linux, Shell, AWS, Azure, Pandas

Interests: Travelling, Basketball, Calisthenics, Robotics, Computer Vision, LLMs, Generative AI, Languages (Spoken)